

2. Structural Load Analysis

Problem: perform a load calculation and analysis for a simple structure (beam, slab, or column)

Consider dead load, live load and wind/earthquake load (basic level).

* present calculations and structural behaviour

Structural Load analysis Report

Submitted by: BVSriarya

Subject: Structural Analysis

Date: 20th June, (2026).

Introduction: ①

Structural load analysis is used to determine the load acting on a structure and understand its behaviour under different loading conditions. In this report, a simply supported reinforced concrete (RCC) beam is analyzed. Consider,

1. Dead load, 2. Live load (LL), 3. Wind load (WL) - Basic

The calculations include total load, support reactions, bending moment, and structural behaviour.

Given data: 2

Structure: Simply supported RCC beam

Beam length (L): 4m

Beam size = 230mm x 450mm

unit weight of RCC = 25 kN/m³

Live load = 5 kN/m

Basic wind load = 1.5 kN/m

Dead load Calculation: 3

Dead load is self weight of the beam.

Formula:

Dead Load = Volume (V) $\times$ Density (ρ)

Step 1: Beam volume per meter length

Width = 230 mm = 0.23 m

Depth = 450 mm = 0.45 m

Volume per meter: $= 0.23 \times 0.45 \times 1$
 $= 0.1035 \text{ m}^3$

Step 2: Dead load

Dead load = $0.1035 \times 25 = 2.59 \text{ kN/m}$

B) Live load calculation

Live load represents movable loads such as people furniture and equipment

Given live load = 5 kN/m .

C) Wind Load calculation (Basic level)

Wind load acts horizontally on structures

Given wind load = 1.5 kN/m

D) Total Load on Beam

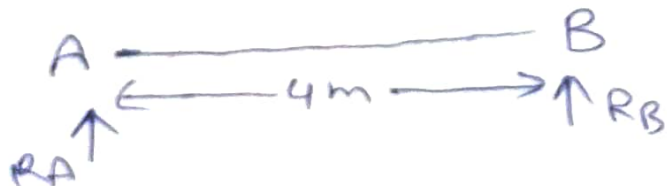
Formula: Total load = Dead load + Live Load
+ Wind load

$= 2.59 + 5 + 1.5$

$= 9.09 \text{ kN/m}$.

4. Structural Analysis:

For S.S.B Carrying uniformly distributed load.



UDL = 9.09 kN/m

Support reactions: $R_A = R_B = \frac{WL}{2}$

Where, $W = 9.09 \text{ kN/m}$, $L = 4\text{m}$

$$R_A = R_B = (9.09 \times 4) / 2$$

$$= \frac{36.36}{2}$$

$$= 18.18 \text{ kN}$$

$$\therefore R_A = 18.18 \text{ kN}; R_B = 18.18 \text{ kN}$$

Maximum Bending Moment:

Formula:

$$M_{\max} = \frac{wl^2}{8}$$

$$= (9.09 \times (4)^2) / 8$$

$$= (9.09 \times 16) / 8$$

$$= \frac{145.44}{8}$$

$$= 18.18 \text{ kNm}$$

5. Structural behaviour:

under dead load:

- Beam bends downward due to self weight.
- compression develops at top
- Tension develops at bottom

under live load:

- Deflection & internal stress increases.
- Beam must safely resist temporary loads.

under wind load:

- Horizontal forces act on the structure
- slight additional stresses develop.
- proper reinforcement stability improves.

Overall behaviour.

* The beam remains stable because, Reaction balance external loads.

* Load transfer occurs from beam to support.

6. Conclusion.

A simply supported RCC beam has analyzed under dead load, live load, and wind load.

Result obtained.

<u>Parameter</u>	<u>Value</u>
Dead load	2.59 kN/m
Live load	5 kN/m
Wind load	1.5 kN/m
Total load	9.59 kN/m
Support reaction	18.18 kN/m
Max. Bending moment	18.18 kN/m

The Beam shows safe structural behaviour under applied loading condition.